NIT3122 Lab Assessment 3 (20%)
Network Design and Implementation

Lab Assessment 3 Requirements:

You are given time **until Friday 11:30pm** to complete Lab Assessment 3 on the **provided Lab platform independently** without assistance from other people including your Lab Tutor and classmates:

(1) Create a new word document called **Lab_Assessment_3_YourName_YourID.docx** where YourName is your real name and YourID is your VU Student ID.
(2) You are required to submit your documentation that provides the network design and describes the steps taken and the screenshots of the system. **All screenshots must show date and time from VM task bar**.
(3) The description of steps does not need to be as detailed as in the lab manual, but it should provide enough information to understand what you have achieved the desired outcome. Mark will be given to screenshot of the system and **quality** of the step description.
(4) Upon completion, upload the following file via dropbox on VU Collaborate:

Lab_Assessment_3_YourName_YourID.docx

Late Submission Penalty

- < 1 day 20 points
- 1-3 days – 30 points each day
- After 3 days – **0** (**zero**) mark

Deduction

- Missing figure numbers/captions – 5 points each
- **Captured screens without task bar – up to 50 points**

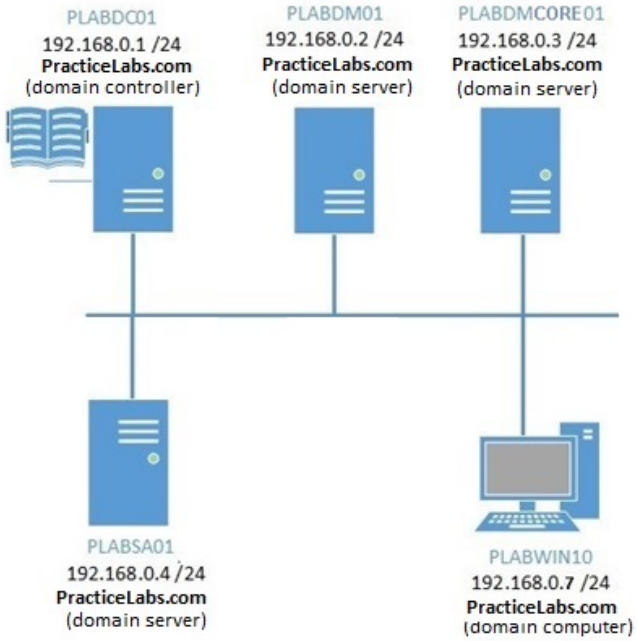
Connecting to your Lab

In this module, you apply "Initial Config" checkpoint before you turn on the virtual machines to carry out the steps.

Important: When you start the lab, **PLABDC01** must be powered on first. Due to shortage on storage space, keep unused VMs **turned off**. Please note that some network services require Active Directory in order to function.

- **PLABDC01** (Windows Server 2016 - Domain Controller)
- **PLABDM01** (Windows Server 2016 - Domain Member Server)
- **PLABDCORE01** (Windows Server 2016 - Domain Member Server Core)
- **PLABSA01** (Windows Server 2016 - Domain Member Server)
- **PLABWIN10** (Windows 10 Enterprise - Domain Workstation)

Lab Diagram



Don't design your network until you finish reading the entire requirements carefully.

Scenario

- You are a network specialist for **PracticeLabs.com**. Your company have four (4) servers running Window Server 2016 (**PLABDC01**, **PLABDM01**, **PLABDMCORE01**, and **PLABSA01**). You are required to design and configure a network with active directory according to the following requirements:
- The root Domain Controller (**PLABDC01**) is located in **Sydney**. Three other server computers must act as active directory domain controller for the **PracticeLabs.com**, to support the high-volume usage of your company. Among them, one should be configured as **Read Only** Domain Controller which does not allow modification of the active directory database. Your company opens one branch in **New York** and takes the child domain name as **NYCYourID**.PracticeLabs.com where **YourID** is your VU Student seven-digit ID without s. Another branch is located in **Shanghai** and has its domain child name called **PVGYourID**.PracticeLabs.com. There is an additional testing computer (**PLABWIN10**) currently configured as member computer of PracticeLabs.com.

The Lab Assessment 3 includes the following three tasks:

Task 1 – Active Directory Design (20%)

- Use commands **date;ipconfig** to record the existing configurations, and take screenshots for **each virtual machine** as well as the top-level Hyper-V Manager.
- Design and draw an **Active Directory Tree** based on the above scenario in your Lab_Assessment_3_YourName_YourID.docx. Clearly label the machine **names**, **IP**, **domain names**, **DNS** IP addresses, and **OU**s.

Task 2 – Active Directory Installation (25%)

- Configure the active directory according to the above design.
- Capture necessary screenshots to demonstrate the steps taken to configure the network, and provide some explanation for your steps.
- After the configuration, take screenshots of **Computers** and **Domain Controllers** folder in Active Directory User and Computers of **PracticeLabs.com**, **NYCYourID**.PracticeLabs.com and **PVGYourID**.PracticeLabs.com to verify the existing domain controllers and member computer in the network.
- Use commands **date;ipconfig** to verify the new configurations, and take screenshots for **each virtual machine** as well as the top-level Hyper-V Manager.

Task 3 – Active Directory Management (55%)

- **A – User Account and Group Management** (15%)
  - Create Organization Units called **LABYourID** and **BLOCKYourID**.
  - Inside **LABYourID** OU, create a new user using **YourName** e.g. **YourFirstName.YourLastName** as Domain **User** who must change password at the first logon.
  - Inside **BLOCKYourID** OU, create a new user using **YourLecturer** e.g. **LecturerFirstName.LecturerLastName** name as **Domain Administrator** whose password never expiries.
  - Take screenshots of the above account creation, **LABYourID** and **BLOCKYourID**.
  - Log on the testing computer to verify the above two users with explanations.

- **B – Software Management** (20%)
  - Create a new GPO named "**NIT3122YourID Software Restrictions**" which don't allow **to use Microsoft Store** for **LABYourID** OU in **PVGYourID**.PracticeLabs.com domain.
  - Use **YourName** to log on the testing computer to verify if **Microsoft Store** can be opened.
  - Then use **YourLecturer** to logon to verify the setting again.
  - Take screenshots of the above procedures and explain the results.

- **C – Application Management** (15%)
  - Create another new GPO named "**NIT3122YourID Computer Restrictions**" which "Don't allow open **Microsoft WORD**" for **PLABWIN10** computer.
  - Use **YourName** to log on the testing computer to verify the setting.
  - Switch to **YourLecturer** to verify the setting again.
  - Take screenshots of the above procedures and explain the results.

- **D – GPO results** (5%)
  Use commands **date;gpresult /r** to take several screenshots of the entire output on screen for **YourName** and **YourLecturer** users. Use commands **date;ipconfig** to take screenshots for the top-level Hyper-V Manager.

Upon completion, log on VU Collaborate and upload via dropbox:

Lab_Assessment_3_YourName_YourID.docx

Please make sure that you won't make any change on Lab Platform after you submit your Lab Assessment 3 on VU Collaborate so that your lecturer could validate your work if necessary.